

RETAILPULSE

Final Analytical Result Report

Retail Transaction Analytics | Python + Power BI

1. Project Overview

RetailPulse is an end-to-end retail analytics project that transforms a large, imperfect transaction dataset into business intelligence. The workflow covers data quality assessment, cleaning, EDA, RFM segmentation, inactivity/churn analysis, demand forecasting and inventory planning, followed by a four-page Power BI dashboard.

2. Final Data Profile

Metric	Observed value
Total records	1,067,371
Total columns	9
Duplicate records	34,335
Missing CustomerID	243,007
Missing Description	4,382
Unique customers	5,942
Unique invoices	53,628
Unique products	5,305
Unique countries	43
Cancellation records	19,494
Negative quantity records	22,950
Zero quantity records	0
Zero price records	6,202
Negative price records	5
Invalid dates	0

3. Customer Analytics Result

RFM analysis evaluates customers through Recency, Frequency and Monetary value. The observed churn output contained 2,989 customers labeled 1 and 2,889 customers labeled 0.

Churn label	Customers
1	2,989
0	2,889

Churn label	Customers
Total	5,878

The 5,878 customers represented in the churn output versus 5,942 unique customers in the broader profiling output should be treated as a modeling/feature-availability difference unless further validation establishes another explanation.

4. Demand Forecasting Result

A 30-day demand forecasting workflow was implemented and prepared for Power BI. The forecast output contains multiple daily rows and supports time-series visualization. Forecast values are analytical estimates and should be evaluated against future actual demand.

5. Inventory Optimization Result

The inventory stage translates demand estimates into average daily demand, lead-time demand, safety stock and reorder-point concepts. Because verified warehouse-stock and supplier-order data are not part of the supplied project results, these outputs are scenario-based planning recommendations rather than actual inventory positions.

6. Power BI Final Dashboard

Page	Purpose	Core visuals
1. Executive Overview	Overall commercial performance	Revenue, orders, customers, products, AOV, monthly trend, country and to
2. Customer Intelligence	Customer value and retention risk	RFM, revenue by segment, inactivity/risk, high-value customers at risk
3. Product & Demand Analysis	Product performance and future demand	Top products, historical demand, 30-day forecast
4. Inventory Optimization	Demand-driven replenishment planning	Forecast, lead-time demand, safety stock, reorder point, recommendations

7. Key Business Insights

- The dataset is large enough for meaningful retail analytics, but data-quality controls are essential before KPI reporting.
- Missing CustomerID values are substantial, so customer conclusions must distinguish identifiable from unattributed transactions.
- Cancellations and negative quantities represent meaningful retail events and require business-aware treatment.
- RFM segmentation supports customer prioritization by value and engagement.
- Churn/inactivity analysis supports targeted retention rather than treating all customers equally.
- Demand forecasting adds a forward-looking layer beyond descriptive reporting.
- Inventory planning connects forecast demand with safety-stock and reorder-point concepts.
- Power BI consolidates descriptive, diagnostic and predictive analysis into a decision-oriented report.

8. Recommended Business Actions

- Prioritize high-value customers with elevated inactivity/churn risk for targeted retention campaigns.
- Monitor high-revenue and high-quantity products for demand changes.
- Use the 30-day forecast as a planning input and review forecast accuracy regularly.
- Review products approaching the modeled reorder point before replenishment decisions.

- Improve source-system controls for CustomerID, descriptions, prices and duplicate transactions.

9. Portfolio Outcome

The project demonstrates an end-to-end Data Analyst workflow: understanding a messy real-world dataset, profiling and cleaning data, producing customer and product analytics, forecasting demand, translating forecasts into inventory planning signals, and communicating results through an interactive Power BI dashboard.

Note: This report distinguishes observed project outputs from assumptions and does not fabricate unverified business outcomes.